

Chuzhditsa: a phonological citation register for Cyrillic

[Author details withheld for double-blind review]*

Abstract

Alphabets only patchily provide a middle layer between everyday spelling and technical transcription: a conventional register for writing a foreign word so that its source pronunciation is recoverable. No major Slavic Cyrillic orthography occupies that layer; loanwords enter through recipient-adapting transcription, with systematic indeterminacy as the result (Bulgarian Вашингтон ~ Уошингтън). We present *Chuzhditsa* (Bulgarian чуждица, ‘foreignism’), a computable Cyrillic-based *phonological citation register*: (i) a compositional grammar of diacritic operations deriving letters for non-Slavic phonemes from attested Cyrillic material; (ii) a prosodic layer for stress, tone, length; (iii) a register-marking typeface whose OpenType rules are the orthography’s rules, executable rather than merely described. Coverage is evaluated against twenty languages under a three-way classification of exact, merged, and declared loss.

Keywords: writing systems, orthography design, Cyrillic, loanword adaptation, phonological transcription, Slavic languages, katakana, biscriptality, OpenType, Bulgarian

1 Introduction

Every literate community borrows words faster than its orthography naturalizes them — an instance of the general problem of fit between a writing system and a changing language (Coulmas 2003). What most orthographies lack is a *productive* middle layer between everyday spelling and technical transcription: an established register in which a foreign word can be written, in the native script, so that a reader can reconstruct its source pronunciation. The layer is not empty — monolingual dictionaries fill it with pronunciation respelling, Japanese with furigana glosses, and the zonal constructed language Interslavic with an extended Cyrillic that already revives the yuses (Merunka 2018) — but no major Slavic Cyrillic orthography has an inventory-*extending*, register-marked device of this kind. Inventory extension for loan phonemes is itself well attested across scripts: Devanagari derives loan-phoneme letters with

*Fonts, build toolchain, and the twenty-language audit are available in an open repository (reference withheld for review); the typeface used for all object-language examples in this manuscript is the system being described.

the nukta dot (*qa, xa, za, fa* from *ka, kha, ja, pha*), Hebrew marks /dʒ ʒ tʃ/ with the geresh (*gimel, zayin, tsadi* + geresh), and Arabic script grew *pe, che, zhe, gaf* for Persian and its neighbors (Daniels & Bright 1996). These attest the diacritic-extension strategy — but each operates inside everyday spelling as an adaptation device; none combines inventory extension with register marking and a citation form held uniform across recipient languages, which is the conjunction proposed here.

Japanese shows the layer can be institutionalized: katakana is a full parallel syllabary whose use *is* the marking of foreignness, and whose adaptation conventions ([nart] → ナイト) are conventionalized enough that, for a given source pronunciation, writers usually converge on one spelling.

Loanwords enter Bulgarian, Russian, Serbian or Ukrainian through recipient-adapting transcription, codified to varying degrees in national norms (Danchev 1982; Gilyarevsky & Starostin 1985; Superanskaya 1978) yet variably applied: sound-based and spelling-based renderings compete (ЛАНЧ ~ ЛЬНЧ for *lunch*), phonemes without Cyrillic letters collapse unpredictably (/θ/ → Т or С; /w/ → В or У), and nothing in the written form signals that a word is a guest rather than a native.

This paper describes Chuzhditsa, a designed orthographic register that supplies the missing device for Cyrillic, piloted on Bulgarian and then generalized to the Slavic superset — the union of the Slavic Cyrillic alphabets.

One distinction must be drawn before anything else, because three different tasks hide under “writing foreign words,” and a system is judged unfairly when measured against a task it does not attempt. *Loanword adaptation* bends a word to the recipient phonology — katakana’s main business, and the business of every integrated borrowing. *Phonemic transcription* records the source phonology in a technical alphabet — the IPA’s business. Between them sits *phonological citation*: writing a foreign word, name, or example so that a native reader can recover an adequate approximation of its source pronunciation, in the native script, without technical training.

Chuzhditsa is primarily the third: unlike katakana it *extends* the recipient inventory to preserve source contrasts rather than collapsing them into native phonotactics; unlike the IPA it stays inside Cyrillic reading habits. Its use as a marking register for not-yet-naturalized loans is a secondary, optional application, and a loan that naturalizes is expected to leave the register.

The system makes three claims of general interest to writing-system design.

First, that a citation register can be built as a *compositional diacritic grammar*: rather than inventing letters *ad libitum*, a small set of recurrent operations — most featurally atomic, one (the hook) relational to its base letter (§4) — derives the entire extension inventory. The analogy to Hangul is one of compositional logic, not of Sampson’s (1985) strict shape-iconicity; systematic diacritic derivation is what Czech, Vietnamese, and the IPA already do, and only

the laryngeal gradient (§4) reaches for genuinely featural letterforms (Sampson 1985; Kim-Renaud 1997). Second, that such a register can be assembled almost entirely from attested material: dead letters of the target orthography’s own history and living letters of related orthographies, with invention reserved for genuine residue. Third, that the register-marking function that Japanese distributes over a second character inventory can instead be carried by a second *typeface register* — realized here as an OpenType family whose substitution (GSUB) and positioning (GPOS) rules are the orthography’s fusion and attachment rules, executable rather than merely described.

2 Background

2.1 Katakana as an adaptation layer

Katakana’s success rests on three properties (Irwin 2011; Stanlaw 2004): conventionalized conversion that renders the loan’s *adapted Japanese pronunciation* — an adaptation that has historically traveled by ear, by dictionary, and sometimes by spelling (Irwin 2011: 76–79) — rather than transliterating the source orthography letter for letter, the orthographic reflex of loanword adaptation as a systematic mapping onto native categories, whether modeled phonologically or perceptually (Kang 2011; Peperkamp & Dupoux 2003); graphic distinctness sufficient to mark register at a glance; and extensibility (ヴ for /v/, ファ for /fa/) when the inventory runs out. Chuzhditsa adopts all three as requirements, and hardens the first: its input is pronunciation only. The idealization should be stated as such — real katakana practice varies by borrowing period, donor language, brand convention, and register (Irwin 2011) — and Chuzhditsa aims to be what katakana is often described as being: the target is a convention strict enough to converge, not a claim that Japanese practice already converges. Katakana also carries a social record that a designed analogue cannot disclaim: loanword marking has been read as a badge of incomplete nativeness, and katakana is deployed to mark foreigners’ accented Japanese as well as foreign words (Irwin 2011; Robertson 2020). We therefore state the register’s semantics narrowly: it asserts that a word’s phonology lies off the native grid — nothing about its right to exist in the language, and nothing about its user. Its proposed semantics is lexical rather than ethnolinguistic — it marks words, never speakers — though a designed intention is not a sociolinguistic guarantee: registers acquire social meanings their designers do not control, and whether readers would honor the boundary is an empirical question for user studies. A naturalized loan is expected to exit the register: fully nativized *tempura* is commonly written 天ぷら, in kanji and hiragana with no katakana trace, while newer loans such as カメラ remain katakana-marked.

2.2 The Cyrillic commonwealth

Cyrillic today serves more than fifty languages, whose principal non-Slavic adaptations Comrie (1996) surveys, and the non-Slavic ones have already engineered letters for most sounds Slavic lacks: Kazakh κ /q/, η /ŋ/, $\mathbb{F}$ (/ʁ/ in Kazakh, reassigned here to /ɣ/: as with $\mathbb{x}$, the borrowing is of the grapheme and its construction); Bashkir $\mathbb{C}$ /θ/, $\mathbb{Z}$ /ð/; Tajik $\bar{\text{и}}$ (final stressed /i/ in Tajik, not length; reassigned here to /i:/) and $\mathbb{x}$ (/h/ in Tajik, reassigned here to /ħ/: the borrowing is of the *grapheme and its compositional construction* — $\mathbb{x}$ plus the place hook — not of its Tajik value, and the Caucasian digraph $\mathbb{x}\mathbb{I}$ is the nearer phonetic precedent); Altai $\ddot{\text{o}}$ /ø/, $\ddot{\text{y}}$ /y/; Belarusian $\ddot{\text{y}}$ /w/; Serbian $\mathbb{I}$ /dʒ/; the Caucasian palochka $\mathbb{I}$. These letters carry decades of typographic and pedagogic practice. A design that imports them inherits that practice for free — an option unavailable to katakana’s designers and, we will argue, the single largest economy in the present system. Modifying Cyrillic is not, however, a politically neutral act: script is a charged identity marker across the region — Serbian Latin/Cyrillic digraphia, the state-driven Latinizations of Kazakh and Moldovan, the recurring Cyrillic-versus-Latin debates — and a proposal that *adds letters* must be read in that light. Chuzhditsa mitigates the charge structurally: it alters no existing orthography, imports only attested material, and petitions no authority (§11, §10). It does not pretend the act is neutral.

2.3 The Glagolitic precedent

Bulgaria is a primary locus of a script replacement: Glagolitic, the first Slavic alphabet, created — or, on Cubberley’s reading, formalized — by Constantine-Cyril for the Moravian mission of 863, was gradually displaced by Cyrillic, devised by Constantine’s disciples at the Preslav Literary School in the 890s on the model of Greek uncial, drawing on Glagolitic mainly for the sounds Greek lacked (Schenker 1995; Cubberley 1996). Glagolitic then survived for a millennium as a special-register script (Croatian liturgical use into the twentieth century; the last *wholly* Glagolitic missal appeared in 1905 (Parčić), and the final printed Glagolitic missal (Vajs 1927) set only the Ordinary of the Mass in Glagolitic beside Latin transliteration). Chuzhditsa knowingly reverses the vector: the visual DNA of the *oldest* Slavic letterforms — round bowls, even low-contrast proportions (the shipped family is two-case, but keeps the unicameral source’s evenness) — marks the *newest* words, and the letters it revives ($\mathbb{x}$ $\mathbb{A}$ $\mathbb{H}$ $\mathbb{I}\mathbb{X}$) inherit the sound-slots of Glagolitic ancestors (the Cyrillic shapes themselves are reworkings). The big yus $\mathbb{x}$ in particular survived in Bulgarian spelling until the 1945 orthographic reform (Comrie & Corbett 1993), by then carrying a non-nasal value, its nasality lost by roughly the eleventh century (Cubberley 1996).

3 Design principles

1. **Determinism, relative to normalized phonological input.** One *specified input* phoneme, one rendering, within a declared mapping: given the same source pronunciation in the reference variety, two transcribers converge. The choice of source pronunciation itself (dialect, variant, spelling pronunciation) is outside the system’s jurisdiction, and loss is permitted but must be declared (§9).
2. **Compositional transparency.** Every diacritic operation encodes one recurrent phonological *relation* to its base letter and applies wherever that relation appears. Some operations are featurally atomic (nasalization, aspiration, length); one is relational rather than atomic — the place hook means “place shifted off the native grid,” interpreted relative to the base and *not* in one articulatory direction (retraction on κ $\times$, advancement on $\mathfrak{z}$, retroflexion on τ Δ $\mathfrak{p}$; §4) — so the operations form a compositional diacritic grammar, featural in Sampson’s extended sense but not in the strict shape-iconic sense Hangul exemplifies; only the laryngeal gradient approaches the latter.
3. **Pair symmetry.** Obstruent letters enter the system in voiceless/voiced pairs aligned with the native series. This principle outranks heritage: the historical theta-letter *fita* (Θ) was rejected because it has no voiced partner and derives from no rule; the Bashkir pair $\mathfrak{c}/\mathfrak{z}$ ($/\theta//\delta/$) does its work systematically.
4. **Precedent first.** Revive native dead letters, then import living Cyrillic letters, and only then derive new combinations; never invent an unattested base shape.
5. **Pan-Cyrillic legibility.** Every base skeleton follows the reader’s existing Cyrillic, and every extension should be guessable at first sight by any Cyrillic reader from its base letter plus a transparent mark — a design *goal*, aimed at rather than achieved, of the kind the orthography-development literature calls ease of transfer (Cahill & Rice 2014). Three properties must be kept distinct here: that a letter exists in Unicode, that it has established use in *some* Cyrillic orthography, and that it will be transparently interpretable to readers of *any* Cyrillic orthography. This paper demonstrates the first two — a Bulgarian reader has no prior acquaintance with Kazakh κ or Bashkir $\mathfrak{z}$, and importing an attested letter buys typographic and pedagogic precedent, not automatic legibility. The third is the criterion the design aims at, and it remains an empirical claim for the reader study of §11.
6. **Register by typeface.** Foreignness is marked by letterform register, not by punctuation or a second inventory: text set in the Chuzhditsa face is thereby marked — though where katakana marks by a stable script-*encoding* contrast, this marks by a *rendering* contrast that a plain-text channel loses (§11); in the terms of the biscriptality literature,

a *diaphasic diglyphia*: functional differentiation between two glyphic variants of one script (Bunčić, Lippert & Rabus 2016; the nearest historical parallel is the German practice of setting foreign words in roman within blackletter text; other register mechanisms — italics, small caps, Japanese ruby glosses — trade distinctiveness against availability and learning cost, and a bespoke face buys the first at the expense of the second two), and a socially meaningful orthographic choice in Sebba’s (2007) sense.

4 The compositional grammar

Table 1 gives the full operation set. The place hook generalizes a precedent internal to the Slavic base inventory itself — graphically, $\mathfrak{w}$ is $\mathfrak{w}$ with a descender (historically a $\mathfrak{w}+\mathfrak{r}$ ligature; Cubberley 1996) — and the glide breve generalizes $\mathfrak{w} \rightarrow \mathfrak{y}$. Where Unicode supplies precomposed hooked letters ($\mathfrak{K} \mathfrak{H} \mathfrak{X} \mathfrak{C} \mathfrak{Z}$) they are used; elsewhere the same hook is written with combining U+0322 ($\mathfrak{T} \mathfrak{A} \mathfrak{P}$).

Mark	Operation	Derivations
◌breve	vowel $\rightarrow$ glide	$\mathfrak{w} \rightarrow \mathfrak{y} \text{ /j/} \cdot \mathfrak{y} \rightarrow \mathfrak{y} \text{ /w/}$
◌hook below	place shifted off the native grid	$\mathfrak{K} \text{ /q/} \cdot \mathfrak{X} \text{ /ħ/} \cdot \mathfrak{H} \text{ /ŋ/} \cdot \mathfrak{C} \text{ /θ/} \cdot \mathfrak{Z} \text{ /ð/} \cdot \mathfrak{T} \text{ /t/} \cdot \mathfrak{A} \text{ /d/} \cdot \mathfrak{P} \text{ /t/}$
◌crossbar	lenition	$\mathfrak{r} \rightarrow \mathfrak{f} \text{ /ɣ/}$
◌double dot	vowel fronting	$\mathfrak{ä} \text{ /æ/} \cdot \mathfrak{ö} \text{ /ø/} \cdot \mathfrak{y} \text{ /y/}$
◌yus tail	nasalization	$\mathfrak{x} \text{ /ɔ̃/} \cdot \mathfrak{a} \text{ /ɛ̃/} \cdot \mathfrak{q} \text{ /ã/} \cdot \mathfrak{ö} \text{ /œ̃/}$
◌ ^h ◌ ^h raised h	aspiration; breathy voice	$\mathfrak{p}^h \mathfrak{t}^h \mathfrak{k}^h \mathfrak{c}^h \mathfrak{b}^h \mathfrak{d}^h \mathfrak{g}^h \mathfrak{z}^h$
◌macron	vowel length	$\mathfrak{ā} \mathfrak{y}$ (precomposed) $\cdot \mathfrak{ā} \mathfrak{ē} \mathfrak{ō}$
l palochka	laryngeal–pharyngeal action	$\mathfrak{r} \mathfrak{k} \mathfrak{r}$ (ejectives) $\cdot$ bare $\mathfrak{l} = \text{ /ʕ/}$
◌b	palatalization	$\mathfrak{h} \mathfrak{b} \text{ /ɸ/} \cdot \mathfrak{h} \mathfrak{b} \text{ /ɸ/} \cdot \mathfrak{x} \mathfrak{b} \text{ /ç/} \cdot \mathfrak{c} \mathfrak{b} \text{ /ç/}$

Table 1: The compositional operations. Each mark encodes a stable operation; most are featurally atomic, while the hook is relational to its base (§4). Precomposed Unicode letters are used where they exist.

Table 2 lays out the complete designed inventory — every letter the system adds to a Slavic Cyrillic base, set in the typeface itself.

4.1 The laryngeal gradient

The three back fricatives that plague Cyrillic transcription of Arabic, English and German — /x/ , /ħ/ , /h/ — receive a designed gradient: articulatory weakening maps to graphic lightening. Native $\mathfrak{x}$ ($\text{ /x/} \sim \text{ /χ/}$) is two full crossing strokes, the densest friction; pharyngeal $\mathfrak{x}$ (/ħ/) is $\mathfrak{x}$ pulled deeper by the place hook; glottal $\mathfrak{h}$ ($\text{ /h/} \sim \text{ /ɦ/}$) is a single open arc, literally the most open glyph of the three; and aspiration $\mathfrak{h}^h$, a feature rather than a segment, is $\mathfrak{h}$ reduced to a superscript. Arabic exhibits the full triple within ordinary vocabulary: $\mathfrak{h} \mathfrak{h} \mathfrak{l} \mathfrak{ā} \mathfrak{l}$ /hila:l/ , $\mathfrak{m} \mathfrak{u} \mathfrak{x} \mathfrak{a} \mathfrak{m} \mathfrak{m} \mathfrak{a} \mathfrak{d}$ /muħammad/ , $\mathfrak{x} \mathfrak{ā} \mathfrak{l} \mathfrak{i} \mathfrak{d}$ /xal:lid/ . We suggest the mapping *articulatory gradient* $\rightarrow$ *visual gradient*

Letter	Unicode	Value	Construction	Origin
Ÿ	U+040E/045E	/w/	y + breve	import: Belarusian
Љ	U+040F/045F	/dʒ/	own shape	import: Serbian
Ҫ	U+04AA/04AB	/θ/	c + hook	import: Bashkir
З	U+0498/0499	/ð/	з + hook	import: Bashkir
Қ	U+049A/049B	/q ɣ/	к + hook	import: Kazakh
Ң	U+04A2/04A3	/ŋ/	н + hook	import: Kazakh
Ҳ	U+04B2/04B3	/ħ/	х + hook	import: Tajik
Ғ	U+0492/0493	/ɣ/	г + crossbar	import: Kazakh (/ʁ/ there; construction borrowed)
Һ	U+04BA/04BB	/h ɦ/	own shape	import: Tatar
Ѕ	U+0405/0455	/dz/	own shape	revival; living in Macedonian (dzelo)
І	U+04C0	/ʎ/; ɾ ɽ ɿ	palochka	import: Caucasus
Ҫ Ҫ̣ Ҫ̤ Ҫ̥	base + U+0322	/t̪ d̪ ɳ t̪/	base + hook	derived by rule
◌ ^h	U+02B0	aspiration	raised miniature Һ	derived
◌ ^h	U+02B1	breathy voice	raised hooked Һ	derived
Ä	U+04D2/04D3	/æ/	a + double dot	import: Mari
Ö	U+04E6/04E7	/ø œ/	o + double dot	import: Altai
Ӳ	U+04F0/04F1	/y ʏ/	y + double dot	import: Altai
Ы	U+042B/044B	/ɨ/; Korean /u/	own shape	import: Russian; also Old Bulgarian
Ү	U+046A/046B	/ʊ ɤ/	nasal o	revival (shape to 1945; nasal value ~11th c.)
Ӱ	U+0466/0467	/ɛ̃/	nasal e	revival: Old Bulgarian
ӱ	U+0468/0469	/jɛ̃/	ligature i+Ӱ	revival: Old Bulgarian
ӱ	U+046C/046D	/jɔ̃/	ligature i+Ӱ	revival: Old Bulgarian
Ҙ Ҙ̣ Ҙ̤ Ҙ̥	any V + U+0328	/ä ẽ ē ī ū œ̃/	vowel + yus tail	derived by rule
Й̄ ӱ̄	U+04E2/3, EE/EF	/i: u:/	vowel + macron	import: Tajik (there ӱ marks final stressed /i/, ӱ̄ is /ø/)
◌̄	U+0304	length	macron	derived
◌̇ ◌̈ ◌̉	U+0301/0300/030C	tone, stress	dictionary layer	derived

Table 2: The complete designed inventory. The letter column is set in the Chuzhditsa face. Every letter is a standard Cyrillic-block codepoint; the combining and modifier characters are likewise standard Unicode (The Unicode Consortium 2024).

as a candidate principle for alphabet extension — a heuristic with a lineage in the featural conscripts, from Bell’s *Visible Speech* (1867), where glyph shape encodes articulation directly, through Shavian and Quikscript, though those aimed at whole-language reform rather than a citation register.

4.2 The Indic stress test

Hindi–Urdu and Bengali cross a four-way laryngeal contrast with a dental/retroflex place contrast. The grammar covers the entire grid with zero new letters: ɾ /t̪/, ɾ^h /t̪ʰ/, ɳ /ɳ̪/, ɳ^h /ɳ̪ʰ/; retroflex ɽ ɽ^h ɳ̪ ɳ̪^h, plus ɣ /ɣ̪/, ɣ /ɣ̪ʰ/, ɣ^h /ɣ̪ʰ/. A single word such as ɽ^hɳ̪k̪ (t̪h̪i:k̪) composes three features on one base — hook, aspiration, length — which is precisely the compositional-grammar claim made concrete.

5 The ligature stratum

Cyrillic has always ligated: ю is a historic ligature of i+oy (the y element later dropped; Cubberley 1996), Serbian њ њ were fused from л/н+ь by Vuk Karadžić in 1818, building on Mrkalj’s digraphs (Comrie & Corbett 1993), and Old Bulgarian possessed iotated nasals. Chuzhditsa’s ligature stratum is therefore conservative, and it is implemented as executable GSUB rules rather than prose: (i) њ њ fuse to њ/љ-forms automatically (нињо, фамиљя set with fused palatals); (ii) the iotated yuses ѡ /jě/ and ѣ /jǝ/ are revived as directly typeable letters, and the sequences ѡ+ѡ and ѣ+ѣ fuse to them automatically (French *bien* → бѡ; Polish *pieć* → пѣч); (iii) ejective digraphs т к’ г л’ ч fuse with the palochka *raised to the upper half*, echoing the apostrophe of scholarly romanization ([t’ k’]) — a fused full-height palochka was rejected in testing because т became confusable with п. The governing rule: only sequences denoting a single phoneme, with Slavic or Old Cyrillic precedent, may fuse; mechanical soft pairs (сь хь) remain visibly two-part.

6 The prosodic stratum

Stress takes the grave of Bulgarian lexicography (бабушка); Mandarin tones 2–4 take acute, caron, grave, with tone 1 unmarked — pinyin minus the macron, which Chuzhditsa reserves for length. The unmarked first tone resolves the collision by construction: mā can only mean a long vowel. Three costs are declared rather than hidden. Tone and stress marks are homoglyphic across source languages, resolved only by knowing the lexical source (katakana shares exactly this property). Diacritic stacking degrades: Vietnamese *phở* wants quality, length and tone at once (phở), and the running-text rule drops length first (phở). And four marks cannot represent six-tone systems without folding (Vietnamese *hỏi/ngã* merge on the caron; *nặng* falls together with *huyền* on the grave). The design verdict generalizes katakana’s own: prosody is optional annotation — a dictionary layer, not an orthographic one — and the system claims no tonal adequacy beyond that annotative role.

7 From Bulgarian to the Slavic superset

The pilot treated Bulgarian’s thirty letters as the base. Nothing in the compositional grammar is Bulgarian-specific, however, and the generalization is administrative rather than structural. We use *superset* in its set-theoretic sense — the union of the Slavic Cyrillic character inventories — and intend no pan-Slavist claim; the older political resonance of a “unifying” Slavic frame is precisely not the argument, as the Ukrainian constraint below makes concrete. The base becomes each language’s own Cyrillic alphabet (for the phoneme inventories, see Comrie & Corbett 1993), and the extension and prosodic strata apply unchanged.

its compositional operations appears to cover the Balkan sprachbund without amendment. The practical beneficiary is not Albanian itself, which is securely Latin-written, but Cyrillic-language text citing Albanian: the *ë* that Macedonian and Serbian renderings currently drop is exactly Ё.

8 Typeface implementation

The Chuzhditsa family — four cuts (the 2b register face, a grotesk, a slab serif, and the Inline citation cut this manuscript is set in), each in Regular, Bold, Italic, and Bold Italic, TTF and CFF/OTF — is generated by a compact parametric builder: every glyph is declared as centerline strokes — lines, arcs, and the full circles that give the face its Glagolitic character — and expanded to outlines with weight and slant as parameters. The construction itself — terminals, joins, the optical-correction canon, fitting, and the small-size behavior of the marks — is the subject of a companion paper (reference withheld for review); here we state only its orthographic consequences.

Capitals stand at 700 units on the design grid, lowercase at an x-height of 500, with true two-case architecture (bowl-and-stem *α*, ascending *η*, descending *ρ*); the Inline cut embedded here is vertically scaled from that grid to match its host serif face. Every combining mark receives computed GPOS mark-to-base anchors (the macron of *ā* attaches roughly 230 units lower than that of *Ā*; the retroflex hook of *ṛ* lands on the letter’s stem), and the ligature stratum is a GSUB *liga* feature — the orthography’s rules are the font’s code, verified by HarfBuzz shaping tests in the build toolchain — a concrete instance of Sproat’s (2000) regularity hypothesis: the mapping from linguistic representation to written form is a regular relation, computable by finite-state means.

The linguistic point of this engineering is falsifiability: an orthographic rule that exists as a GSUB lookup either fires or does not, so the specification’s claims about fusion and attachment are machine-checkable properties of an artifact rather than intentions in prose. The manuscript the reader is holding sets all object-language material in this typeface. Figure 1 shows the register in context, in a Bulgarian sentence, against the two current practices: the bespoke face marks more distinctively than italics and, unlike them, preserves the source contrasts (*ŷ* /*w*/, *ç* /*θ*/, *ǣ* /*æ*/, *н* /*η*/) — at the cost, stated in §11, of depending on font control.

9 Evaluation: the twenty-language audit

Method. The audit set is not a demographic top-twenty: it is a high-speaker-count convenience sample adjusted for phonological diversity and Slavic relevance, comprising English, Mandarin, Hindi–Urdu, Spanish, French, Arabic (MSA), Bengali, Portuguese, Russian, Japanese, German, Korean, Vietnamese, Turkish, Italian, Persian, Swahili, Greek, Polish, and

Прекарахме ѱйкенѿа в планината и казахме ѿѿѿкс. Прекарахме <i>weekenda</i> в планината и казахме <i>thanks</i> . Прекарахме уикенѿа в планината и казахме тенкс.
--

Figure 1: The register in context, at reading size, in a Bulgarian sentence (“We spent the weekend in the mountains and said thanks”). Top: the foreign words set in the Chuzhditsa face inline in the Cyrillic body face — the proposed register. Middle: the current practice of Latin italics. Bottom: plain recipient-adapting transcription, which marks nothing.

Indonesian; clicks were excluded by scope. Each language is audited under one declared reference variety (General American English; Standard Mandarin; Delhi Hindi; Castilian Spanish; Parisian French; MSA; Brazilian Portuguese; Seoul Korean; Hanoi Vietnamese; and the respective standards elsewhere), with inventories taken from the language illustrations of the *Handbook of the IPA* (International Phonetic Association 1999), the *Journal of the International Phonetic Association*’s subsequent Illustration series for declared varieties the Handbook lacks, and standard language descriptions (for Slavic, Comrie & Corbett 1993; for Bengali and Swahili, where no Illustration matches the declared variety, standard descriptions).

Every phoneme of each inventory is classified three ways: *exact* (a rendering that preserves all of that language’s contrasts involving the phoneme), *conventional merger* (a declared many-to-one mapping, enumerated below), or *declared loss* (prosodic folding, plus two segmental cases: length under tone, and /ʔ/ outside Arabic contexts). A rendering that depends on position (/w/, /ɲ/) counts as exact when the positional rule is stated; Spanish is accordingly audited on broad-phonetic input, its spirantization rendered by rule (/b d g/ → Б Δ Γ initially and after nasals; В З F elsewhere). The system is deterministic relative to this input: variety choice is a parameter of the audit, not of the mapping. Against existing practice the relevant baselines are the national transcription traditions — Bulgarian’s prescriptive system (Danchev 1982) and its analogues (Gilyarevsky & Starostin 1985; Superanskaya 1978) — which are recipient-adapting by design; Chuzhditsa differs precisely in preserving source contrasts those systems merge, which is what the citation-register framing predicts.

One boundary of this evaluation must be stated plainly: it is an *engineering coverage audit*, internally scored. The author of the mappings selected the reference varieties, declared the acceptable mergers, assembled the examples, and judged exactness; the machine-readable dataset makes every judgment inspectable and refutable, but no independent phonologist has yet re-scored it, no blinded transcribers have been tested for convergence, and no unseen language has been audited that was not available while the system was being refined (the Albanian application is a stress test, not external validation).

The sample is moreover biased toward segmental inventories: complex lexical tone (only Mandarin and Vietnamese), ATR and vowel-harmony systems, contrastive phonation, and ejectives are under-represented or routed to loss, so the coverage the audit reports is best read as an upper bound. The audit is reported qualitatively — the per-language mergers and losses

of Table 4, plus the full example set — rather than as a summary statistic. And because the author defines which mappings count as conventional mergers, the audit scores representational capacity against the declared rules, not adequacy against an external standard — functional load of the merged contrast, minimal-pair recoverability, or blind-transcriber convergence, the appropriate next instruments. What the audit establishes is representational capacity under the declared rules — not communicative success, which is the next experiment’s question.

The audit’s full 125-item example set is printed as Appendix A and accompanies the repository in machine-readable form (`data/audit_20_languages.json/.csv`, generated from the same source table as the appendix and the per-language guides, so the three cannot drift), together with the phoneme-to-grapheme mapping it instantiates (`data/chuzhditsa.json`).

Table 4 summarizes the audit per language: the reference variety, the extension letters the language requires, the conventional mergers engaged, and the declared losses. The pattern worth reading off is that no language needs more than a small subset of the inventory, that the mergers cluster on rhotics and sibilant places, and that declared loss is confined to prosody and two segmental cases: length under tone, and Persian /ʔ/.

Language	Reference variety	Extensions required	Mergers engaged	Declared losses
English	General American	ŷ ʒ ʒ ä ɳ ɳ ɳ ɳ ɳ ɳ	rhotic → p	—
Mandarin	Standard (Beijing)	ɳ ɳ ɳ tones	/ʈʂ ʈʂ/ → ɳ; /ʒ/ → ʂ	sandhi; T1 = neutral
Hindi–Urdu	Delhi	ɳ ʈ ʈ ɳ ɳ ɳ ɳ ɳ ɳ ɳ	/h/ → ɳ	—
Spanish	Castilian	ʒ ʒ ɳ ɳ	rhotics (/β ð ɣ/ by positional rule)	—
French	Parisian	ɳ ʌ ʌ ʌ ʌ ʌ ʌ ʌ	/ʌ/ → p; /ɛ ɔ œ/ → ɛ ɔ ɔ; /ɥ/ → ɳ	—
Arabic	MSA	ʒ ʒ ɳ ɳ ɳ ɳ ɳ ɳ	/χ/ → ɳ; emphatics → plain	—
Bengali	Standard (Kolkata)	ɳ ʈ ɳ ɳ ɳ ɳ ɳ ɳ	/ɔ/ → ɔ	—
Portuguese	Brazilian	ɳ ɳ ɳ ɳ ɳ	/ʌ/ → p; /ɛ ɔ/ → ɛ ɔ	—
Russian	Standard	ɳ (native)	/ʂ/ → ɳ ɳ	reduction (phonemic input)
Japanese	Tokyo	ɳ ɳ ɳ	/ɸ/ → ɸ; /w/ → ɳ; /z/ → ɳ; /n/ → ɳ	pitch accent
German	Standard	ö ʏ ɳ ɳ ɳ ɳ ɳ	/ɛ:/ → ɛ; /e:/ → ɛ	—
Korean	Seoul	ɳ ɳ ɳ ɳ	tense = geminate	—
Vietnamese	Hanoi	ɳ ɳ ɳ ɳ ɳ ɳ ɳ	/b d/ → ɳ ɳ	6 tones → 4; length/tone
Turkish	Istanbul	ö ʏ ɳ ɳ ɳ	/ɰ/ → ɳ	—
Italian	Standard	ɳ	/ɛ ɔ/ → ɛ ɔ; /ʌ/ → ɳ	—
Persian	Tehran	ɳ ɳ ɳ ɳ	/ʒ/ → ɳ	/ʔ/ dropped
Swahili	Standard	ʒ ʒ ɳ ɳ ɳ	implosives → ɳ ɳ	—
Greek	Standard Modern	ʒ ʒ ɳ	—	—
Polish	Standard	ɳ ʌ ʌ ʌ ɳ	/ʒ/ → ɳ; /ʈʂ/ → ɳ	—
Indonesian	Standard	ɳ ɳ ɳ ɳ	—	—

Table 4: Per-language audit summary. “Extensions required” lists the letters and marks the language’s inventory requires; palatalization via ɳ and length via ɳ are counted where load-bearing. Mergers and losses are the declared ones of §9; the appendix examples exercise a subset of each row.

Table 5 samples the audit; the full 125-item set is Appendix A. Declared mergers are: all rhotics → p; /ʒ//ʒ/ → ɳ; /ɸ//f/ → ɸ; /h//h/ → ɳ; implosives → ɳ ɳ; /q//ʒ/ → ɳ (so Persian ʒ and ʒ, one phoneme in the Tehran reference, both → ɳ); the pharyngealized emphatics /tʂ dʂ sʂ ðʂ/ → plain ɳ ɳ ɳ ɳ (Arabic); the open/close mid vowels /e//ɛ/ → ɛ, /o//ɔ/ → ɔ, and

/ø//œ/ → ö (Romance, Bengali); Korean tense (fortis) consonants written as geminates (one analysis of the contrast); /x//χ/ → x; the affricates /tʃ tʂ tɕ/ → ʈ and /dʒ dʒ dʒ/ → ɽ, with /ʒ//ʒ/ → ʒ (so Polish *cz/ć* and Mandarin’s retroflex/palatal series each fall together); Russian /ɕ:/ → ɕ; Japanese /ʑ/ → ɕ (the fricative falling in with the affricate series), /ɲ/ → ɲ, and /ɕ/ → ɕ by entrenchment (суши); Italian /ʎ/ → ʎ (degemination); /v//β/ → β; /l//ʎ/ → ʎ; French /ɥ/ → ɥ (the glide written with its vowel counterpart, licensed by the systematic /y/~[ɥ] alternation); Japanese /w/ ([ɰʷ], compressed-rounded) → ɰ by convention, where Korean /w/ → ɰ; German /ɛ://e:/ → ē; English input referenced to a rhotic accent (GA), with tense /i//u/ written ī/ū although General American has no phonemic length — a transcription convention imported for reader familiarity, flagged rather than hidden. /ʁ/ is written (ʁ); /ʀ/ is written with the apostrophe ‘ in Arabic contexts and dropped elsewhere, as the shipped mapping (data/chuzhditsa.json) specifies. Declared losses are prosodic, with two segmental cases: Mandarin Tone 1 and the neutral tone are both unmarked and so coincide; six-tone systems fold into four marks (Vietnamese *hỏi/ngã* merge on the caron, and *nặng* falls together with *huyền* on the grave); Vietnamese phonemic vowel length is neutralized wherever a tone mark is present (length drops under tone); and /ʀ/ is dropped outside Arabic contexts (Persian). A rendering that engages none of these is *exact* in the Method’s sense.

Source		IPA	Chuzhditsa
English	<i>thanks</i>	[θæŋks]	çăŋкс
English	<i>weekend</i>	[ˈwi:kend]	ўйкенд
Mandarin	北京 <i>Běijīng</i>	[peɪ.tɕiŋ], T3+T1	Пёйчин
Hindi	थीक <i>thīk</i>	[tʰi:k]	ṭʰīk
Arabic	محمد <i>Muhammad</i>	[muˈħammad]	Мухаммад
French	<i>croissant</i>	[kʁwasɑ̃]	крўасɑ
French	<i>bien</i>	[bjɛ̃]	бѣ
Polish	<i>Wrocław</i>	[ˈvrɔtswaf]	Вроцѣф
Japanese	東京 <i>Tōkyō</i>	[to:kʰo:]	Тōкьō
Korean	한글 <i>hangul</i>	[hangwɭ]	һангыл
Georgian	წყალი <i>c’qali</i>	[ts’q’ali]	цқ али
Albanian	<i>Shqipëria</i>	[ʃcipəˈria]	Шкьипьриа

Table 5: Sampled audit items, plus two out-of-sample demonstrations (the Georgian ejectives and the Albanian stress test); the Chuzhditsa column is set in the typeface under discussion, with live GSUB fusion and GPOS anchoring. In the Georgian row цқ fuses while қ|, outside the declared fusion set of the ligature stratum, sets as a plain sequence by design. In the Mandarin row the caron on ě is tone 3, attached to the first syllable’s vowel; tone 1 is unmarked by rule.

10 Adoption: lessons from planned and organic systems

A designed register lives or dies socially, not technically; the sociolinguistics of script choice behaves like the sociolinguistics of language choice (Fishman 1977; Unseth 2005). Six his-

torical cases bound the space Chuzhditsa enters. The comparison is organized by recurrent factors rather than narrative sympathy: community ownership, institutional sponsorship, literacy burden, identity payoff, compatibility with existing orthographic practice, domain of use, prestige and stigma, and the mundane infrastructure of input methods and type.

The planned failures. Volapük collapsed within a decade, less on linguistic defects than on its inventor’s proprietary control of the standard (Garvía 2015). Esperanto — maximally learnable, energetically organized — plateaued as a voluntary subculture: it extended nobody’s existing practice and attached to no prior community, so adopting it meant *joining a new identity* rather than exercising an old one (Forster 1982; Garvía 2015).

The Deseret alphabet is the sharpest lesson, because it controls for institutional power: commissioned by Brigham Young, taught in classes from church-printed primers, backed by church funds, it was abandoned within roughly two decades — sponsorship without a felt need or an identity payoff bought nothing (Alder, Goodfellow & Watt 1984). Learnability and patronage, the two levers a designer can pull, decided none of these cases.

Nearer to Chuzhditsa than any whole-language project is the Initial Teaching Alphabet, a transitional English orthography whose central difficulty — treated at length by its own champions (Pitman & St John 1969) — was the dual system and the transition back to standard spelling, and which was largely abandoned in the following decade: the same leave-the-register burden a citation register asks its users to carry when a loan naturalizes.

The designed success. Among these cases, Hangul is the engineered script that won, and its path is instructive precisely because the design was not what won. Completed in 1443 and promulgated in 1446 (Ledyard 1998), it was opposed at once by the literati — the collective Ch’oe Malli memorial of 1444 — and spent four and a half centuries outside prestige writing, surviving in low-stakes registers: women’s correspondence, vernacular fiction, private devotion (Lee & Ramsey 2011). What revalued it was not its featural elegance but nineteenth-century nationalism, culminating in the 1894 Kabo edict that made the national script the base of official documents (King 1998). The mandate arrived 448 years after the design; the niche did the surviving.

The organic success. Katakana never asked anyone to change anything. It began as monks’ abbreviated annotation glosses on Chinese texts in the early Heian period, drifted into a division of labor with hiragana over a millennium, and was codified only in 1900 by the implementing regulations of the Elementary School Order — with the loanword-register specialization consolidating later still (Seeley 1991; Twine 1991). Codification ratified practice; it did not create it.

The pattern reduces to four conditions, against which Chuzhditsa can be scored honestly. (i) *Additive, not substitutive*: the winners changed no existing spelling; Chuzhditsa alters nothing in any native orthography by construction. (ii) *Identity attachment*: Hangul survived on Korean-ness, not on design merit; Chuzhditsa’s revivals — the yuses, the Glagolitic let-

transforms — are the analogous move available to a designed system, heritage rather than invention. (iii) *A survivable low-stakes niche before any mandate*: Hangul had the vernacular genres, katakana the annotation margin; the dictionary pronunciation field, the language classroom, and the subtitle are proposed as the equivalent low-stakes niche. (iv) *Codification last*: every mandate that worked ratified existing practice, which is why this specification petitions no orthographic authority.

Six cases cannot license induction, and none of this predicts uptake; the comparison identifies necessary-looking conditions, not sufficient ones. One condition, moreover, cannot be engineered around: on the planned/organic axis Chuzhditsa sits with Esperanto and Deseret, not with katakana. Assembly from attested material is a mitigation — every letter arrives with a history already attached — but it is not an exemption, and the Deseret case fixes an upper bound on what design quality plus sponsorship can buy without a community that recognizes itself in the script.

11 Limitations

Register versus encoding. The register’s marking lives at two levels that must not be conflated. The phonological extensions are plain-text-safe: they are standard Unicode letters that survive copy, paste, search, and font substitution. The *visual* register, however, is carried by typeface choice — a diaphasic diglyphia in Bunčić’s terms, not an encoded script contrast — and it degrades exactly where font control degrades: plain-text channels keep the letters but lose the look. The register is moreover invisible to non-visual access — a screen reader or a plain-text client receives the phonological letters but no foreignness signal at all, a cost of marking by font rather than by inventory. Whether readers actually perceive the Glagolitic-derived forms as “foreign register” rather than “archaic,” “ecclesiastical,” or merely “a display font” is an empirical question this paper does not answer; a perception study — mixed native/Chuzhditsa text, register-identification and readability tasks with Cyrillic readers — is the single most valuable next experiment.

Input assumptions. The mapping consumes a source pronunciation in a declared reference variety. Dialect choice, phonemic versus phonetic input, and spelling-influenced variant pronunciations are parameters the user supplies, not problems the system solves; learnability of the extension letters, and the cost of input methods for them, are likewise open. The keyboard prototype in the repository addresses input mechanics but not habit.

System-internal residue. The place hook is polysemous by direction — uvular on κ , pharyngeal on χ , *fronting* to interdental on $\mathfrak{c}$ $\mathfrak{z}$, retroflex on τ Δ $\mathfrak{p}$ — so “retraction” would misname it; the mark encodes “off the native grid,” not one direction. No attested language contrasts two of its values on one base letter, so the ambiguity is theoretical, but it is a real weakening of the one-mark-one-meaning claim. / $\mathfrak{c}$ / keeps two spellings ($\mathfrak{c}\mathfrak{b}$ under contrast, $\mathfrak{w}$

by entrenchment) — the one point where usage beat system. Six-tone prosodies fold lossily into four marks. Click phonology is future work, as is a mark-to-mark (mkmk) positioning pass for the rare double-stacked vowel.

12 Conclusion

What this paper demonstrates is deliberately narrower than what it proposes. It demonstrates that a coherent, technically implementable candidate for a katakana-like citation register can be *constructed* on an alphabet without inventing a parallel inventory: a compositional grammar over attested Cyrillic material covers twenty phonologies under internally scored audit, one historical script supplies the register aesthetics, and the whole specification compiles to a font whose OpenType rules *are* the orthography. Whether the register *communicates* — whether readers recover pronunciations, transcribers converge, and the typeface signals “foreign citation” rather than “archaic” — is not demonstrated here; those are the claims the perception study of §11 exists to test, and until it is run this paper should be read as a design proposition with an inspectable artifact, not as evidence of communicative success. The generalization from Bulgarian to the Slavic superset cost one letterform (e) and no grammar changes, which we take as evidence that the operation layer, not the letter list, is the right unit of design.

The contribution we would defend most broadly is the vertical integration. A phonological distinction here is traceable, without a gap, through a graphemic operation, a Unicode sequence, a glyph construction, an OpenType behavior, and a rendered artifact — and each level is testable at its own altitude: mappings against inventories, spellings against shaping, constructions against binaries. A companion paper (reference withheld for review) documents the construction level; the same end-to-end coherence is what makes every specification-level claim in this paper — mapping, fusion, attachment — falsifiable by mechanical means, and it is a property worth wanting in any designed writing system, whatever the fate of this one.

References

- [1] Alder, D. D., P. J. Goodfellow & R. G. Watt. 1984. Creating a new alphabet for Zion: The origin of the Deseret alphabet. *Utah Historical Quarterly* 52(3). 275–286.
- [2] Bell, Alexander Melville. 1867. *Visible speech: The science of universal alphabets*. London: Simpkin, Marshall.
- [3] Bunčić, Daniel, Sandra L. Lippert & Achim Rabus (eds.). 2016. *Biscriptality: A sociolinguistic typology*. Heidelberg: Universitätsverlag Winter.

- [4] Cahill, Michael & Keren Rice (eds.). 2014. *Developing orthographies for unwritten languages*. Dallas, TX: SIL International.
- [5] Comrie, Bernard. 1996. Adaptations of the Cyrillic alphabet. In Peter T. Daniels & William Bright (eds.), *The world's writing systems*, 700–726. New York: Oxford University Press.
- [6] Comrie, Bernard & Greville G. Corbett (eds.). 1993. *The Slavonic languages*. London: Routledge.
- [7] Coulmas, Florian. 2003. *Writing systems: An introduction to their linguistic analysis*. Cambridge: Cambridge University Press.
- [8] Cubberley, P. 1996. The Slavic alphabets. In Peter T. Daniels & William Bright (eds.), *The world's writing systems*, 346–355. New York: Oxford University Press.
- [9] Danchev, Andrei. 1982. *Българска транскрипция на английски имена* [Bulgarian transcription of English names], 2nd edn. Sofia: Narodna prosveta.
- [10] Daniels, Peter T. & William Bright (eds.). 1996. *The world's writing systems*. New York: Oxford University Press.
- [11] Fishman, Joshua A. (ed.). 1977. *Advances in the creation and revision of writing systems*. The Hague: Mouton.
- [12] Forster, Peter G. 1982. *The Esperanto movement*. The Hague: Mouton.
- [13] Garvía, Roberto. 2015. *Esperanto and its rivals: The struggle for an international language*. Philadelphia: University of Pennsylvania Press.
- [14] Gilyarevsky, R. S. & B. A. Starostin. 1985. *Иностранные имена и названия в русском тексте* [Foreign names and titles in Russian text], 3rd edn. Moscow: Vysshaya shkola.
- [15] International Phonetic Association. 1999. *Handbook of the International Phonetic Association: A guide to the use of the International Phonetic Alphabet*. Cambridge: Cambridge University Press.
- [16] Irwin, M. 2011. *Loanwords in Japanese*. Amsterdam/Philadelphia: John Benjamins. doi:10.1075/slcs.125
- [17] Kang, Yoonjung. 2011. Loanword phonology. In Marc van Oostendorp, Colin J. Ewen, Elizabeth Hume & Keren Rice (eds.), *The Blackwell companion to phonology*, vol. 4, 2258–2282. Malden, MA: Wiley-Blackwell.

- [18] Kim-Renaud, Young-Key (ed.). 1997. *The Korean alphabet: Its history and structure*. Honolulu: University of Hawai‘i Press.
- [19] King, Ross. 1998. Nationalism and language reform in Korea: The questione della lingua in precolonial Korea. In Hyung Il Pai & Timothy R. Tangherlini (eds.), *Nationalism and the construction of Korean identity*, 33–72. Berkeley: Institute of East Asian Studies.
- [20] Ledyard, Gari K. 1998. *The Korean language reform of 1446*. Seoul: Sin’gu Munhwasa.
- [21] Lee, Ki-Moon & S. Robert Ramsey. 2011. *A history of the Korean language*. Cambridge: Cambridge University Press.
- [22] Merunka, Vojtěch. 2018. *Interslavic zonal constructed language: An introduction for English-speakers*. Prague: Slovanská unie.
- [23] Parčić, Dragutin Antun (ed.). 1905. *Rimski misal slověnskijm jezikom* (Missale Romanum slavonico idiomate), 3rd edn. Rome: Typis S. Congregationis de Propaganda Fide.
- [24] Peperkamp, Sharon & Emmanuel Dupoux. 2003. Reinterpreting loanword adaptations: The role of perception. In Maria-Josep Solé, Daniel Recasens & Joaquín Romero (eds.), *Proceedings of the 15th International Congress of Phonetic Sciences*, 367–370. Adelaide: Causal Productions.
- [25] Pitman, James & John St John. 1969. *Alphabets and reading: The Initial Teaching Alphabet*. London: Pitman.
- [26] Robertson, Wesley C. 2020. *Scripting Japan: Orthography, variation, and the creation of meaning in written Japanese*. London: Routledge.
- [27] Sampson, Geoffrey. 1985. *Writing systems: A linguistic introduction*. Stanford: Stanford University Press.
- [28] Schenker, Alexander M. 1995. *The dawn of Slavic: An introduction to Slavic philology*. New Haven: Yale University Press.
- [29] Sebba, Mark. 2007. *Spelling and society: The culture and politics of orthography around the world*. Cambridge: Cambridge University Press.
- [30] Seeley, Christopher. 1991. *A history of writing in Japan*. Leiden: Brill.
- [31] Sproat, Richard. 2000. *A computational theory of writing systems*. Cambridge: Cambridge University Press.

- [32] Stanlaw, James. 2004. *Japanese English: Language and culture contact*. Hong Kong: Hong Kong University Press.
- [33] Superanskaya, A. V. 1978. *Теоретические основы практической транскрипции* [Theoretical foundations of practical transcription]. Moscow: Nauka.
- [34] Twine, Nanette. 1991. *Language and the modern state: The reform of written Japanese*. London: Routledge.
- [35] The Unicode Consortium. 2024. *The Unicode Standard*, Version 16.0.0. South San Francisco, CA: The Unicode Consortium.
- [36] Unseth, Peter. 2005. Sociolinguistic parallels between choosing scripts and languages. *Written Language & Literacy* 8(1). 19–42. doi:10.1075/wll.8.1.02uns
- [37] Vajs, Josef (ed.). 1927. *Rimski misal slověnskimi jezikom* (Missale Romanum slavonico idiomate). Rome: Typis Polyglottis Vaticanis.

A The twenty-language audit: full example set

The complete audit set behind Table 4 and Table 5, printed as Table A1: 125 examples across the twenty audit languages, each with source form, source-variety IPA, Chuzhditsa rendering (set live in the typeface, with GSUB fusion and GPOS anchoring), and the extension letters the rendering exercises noted where instructive. Language headers restate the declared reference variety and the declared mergers and losses of §9; every rendering not engaging a declared merger or loss is *exact* in the sense of the Method paragraph of §9. This appendix is generated from the same machine-readable table (`data/audit_20_languages.json`) distributed with the fonts.

Table A1: The full 125-item audit example set.

Source	IPA	Chuzhditsa	Notes
English (General American) — mergers: rhotic → p			
weekend	/ˈwi:kɛnd/	ЎЙКЕНД	ў /w/, ъ long /i:/
thanks	/θæŋks/	ЇӘҢКС	џ /θ/, ә /æ/, Ң /ŋ/ — three new letters
this	/ðɪs/	ЗИС	з /ð/
thriller	/ˈθrɪlɹ-/	ЇРИЛЪР	џ + ъ for /ə/
birthday	/ˈbɜθdeɪ/	БЪРЇДЕЙ	ъ /ɜ/, џ /θ/
world	/wɜːld/	ЎЪРЛА	ў + ъp
squirrel	/ˈskwɜːəl/	СКЎЪРЪЛ	ў in a cluster

Source	IPA	Chuzhditsa	Notes
Harry	/ˈhæri/	Һәри	h /h/, ä /æ/
ring	/rɪŋ/	рин	ң — no fake r
cat · cut · cart	/kæt/ /kʌt/ /kart/	кәт · кьт · карт	minimal triple preserved

Mandarin (Standard (Beijing)) — mergers: tʂ tɕ → ч; ʐ → ж; declared loss: tone sandhi; T1 = neutral tone

北京 Běijīng	/pèi.tɕíŋ/	Пěйчин	unaspirated b/j are voiceless → п/ч; tone 3 ǚ, ы
茶 chá	/tʂʰǎ/	чʰǎ	aspiration ʰ + tone 2
谢谢 xièxie	/ɕjê.ɕje/	сьёсьё	ɕ → сь
人 rén	/zǎn/	жън	ʐ → ж, ь /ə/
上海 Shànghǎi	/ʂâŋ.xài/	Шàнхǎй	ʂ → ш, x /x/
猫 māo	/máu/	мао	tone 1 = unmarked
气 qì	/tɕʰí/	чʰй	aspirated affricate + tone 4

Hindi-Urdu (Delhi) — mergers: ɦ → h

भारत Bhārat	/ˈbʱaːrət/	Бʱārьт	ʱ breathy voice, ā length, ь /ə/
धन्यवाद dhanyavād	/dʱənjəˈuaːd/	Дʱъньйъвāд	дʱ, two schwas → ь
ठीक ṭhīk	/tʰiːk/	тʰйк	retroflex ɽ + aspiration + length
दिल्ली Dillī	/d̪illiː/	Диллй	dental д, no hook; geminate
घर ghar	/gʱər/	гʱьр	гʱ breathy r
झील jhīl	/dʒʱiːl/	цʰйл	ɟʱ — voiced affricate with breathiness
चाय chāy	/tʃaːj/	чāй	ч
पानी pānī	/paːniː/	пāнй	double length
लड़का lar̥kā	/ləɽkaː/	лъркā	ɽ /ɽ/ retroflex flap
हवा havā	/həˈuaː/	һъвā	h /h/ (merges with /h/)

Spanish (Castilian) — mergers: rhotics → p (β ð ɣ by positional rule)

cerveza	/θerˈβeθa/	ҫервеҫа	Castilian /θ/ → ҫ
jamón	/xaˈmon/	хамон	j = /x/ → native x
agua	/ˈaɣwa/	аг ъа	ɣ /ɣ/ + ь /w/
niño	/ˈniɲo/	нињо	ɲ → нь (native mechanism)
Madrid	/maˈðrið/	Мазриз	/ð/ → з
paella	/paˈeʎa/	паеља	ʎ → лъ

French (Parisian) — mergers: ɥ → p; ɛ ɔ œ → е о ө; ɥ → ь

bonjour	/bɔ̃ʒur/	Бхжур	ж /ɔ̃/ — the resurrected yus
---------	----------	-------	------------------------------

Source	IPA	Chuzhditsa	Notes
vin	/vɛ̃/	ВА	А /ɛ̃/
blanc	/blā/	БЛҢ	Ң /ā/ — the yus tail, by rule
croissant	/kʁwasā/	кр҃ѡсҢ	҃ + Ң
deux	/dø/	ДӖ	Ӗ /ø/
rue	/ʁy/	р҃ѣ	҃ /y/
parfum	/pɑʁfœ̃/	парфӖ	Ӗ /œ̃/ — umlaut + nasal, two marks
bien	/bjɛ̃/	БҢ	Ң /jɛ̃/ — the iotated yus, a ligature
grenouille	/gʁənuij/	грѣнуѣ	schwa → ѣ

Arabic (MSA) (Modern Standard Arabic) — mergers: ځ → x; emphatics → plain

هلال hilāl	/hiˈlaːl/	Һилāl	h /h/ — light h
محمد Muḥammad	/muˈħammad/	Мухаммад	ħ /ħ/ — pharyngeal h
خالد Khālīd	/ˈxaːlid/	Хālид	x /x/ — the native x: all three H's in three names
قطر Qaṭar	/ˈqatˤar/	Қатар	қ /q/ uvular
غزال ghazāl	/ɣaˈzaːl/	Ғазāl	ғ /ɣ/
عمر ʿUmar	/ˈʕumar/	Ўмар	Ў /ʕ/ ayn
شكرا shukran	/ˈʃukran/	шукран	native letters only

Bengali (Standard (Kolkata)) — mergers: ঔ → o

ঢাকা Ḍhākā	/d̪ʱaka/	Д̪ʱака	retroflex + breathy voice at once
ঠাকুর Ṭhākur	/t̪ʱakur/	Т̪ʱакур	Tagore, spelled as he sounds
কলকাতা Kolkātā	/kolkat̪a/	Колката	no new letters
বড় bôṛo	/bɔ̃ɽo/	Борѡ	р /ɽ/
মিষ্টি misht̪i	/mɪʃt̪i/	мишті	ш + ʈ in a cluster
ভাই bhāi	/bʱai/	Бʱай	бʱ
মাছ māchh	/mat̪ʰ/	мачʰ	final aspiration
ধন্যবাদ dhonnobād	/d̪ʱɔ̃nnobad̪/	Д̪ʱоннобад	дʱ + geminate

Portuguese (Brazilian) — mergers: ɤ → p; ε ɔ → e o

São Paulo	/sẽw̃ ˈpawlu/	Сау Паулу	ґ nasal + y semivowel
pão	/pẽw̃/	пау	nasal diphthong
coração	/koraˈsẽw̃/	корасґу	-ção → -çґу, systematically
obrigado	/obriˈgadu/	ОБРИГАДУ	no new letters
manhã	/mẽˈɲẽ/	манґя	нь + nasal via я

Russian (Standard) — mergers: Ѣ → шч; declared loss: reduction (phonemic input)

мышь	/mɨʂ/	мыш	ы comes back home; no silent ь
Хрущёв	/xɾʊˈʦːɐf/	Хрушчѡв	Rus. щ /ʦː/ ≠ Bul. щ /ʃt/

Source	IPA	Chuzhditsa	Notes
бабушка	/ˈbabuʂkə/	БАБУШКА	phonemic, not phonetic: reduction goes unwritten
ёлка	/ˈjɐlkə/	ЙОЛКА	ё → йо
Пётр	/pʲɐtr/	ПЬОТР	palatalization with ь

Japanese (Tokyo) — mergers: $\phi \rightarrow \Phi$; $\omega \rightarrow y$ (conv.); $\mathfrak{z} \rightarrow \mathfrak{z}$; $\mathfrak{n} \rightarrow \mathfrak{n}$; $\mathfrak{c} \rightarrow \mathfrak{c}$ (entrenched); declared loss: pitch accent

東京 Tōkyō	/to:kʰo:/	ТОКЬО	double length ̄
大阪 Ōsaka	/o:saka/	ОСАКА	initial long vowel
可愛い kawaii	/kaɕai:/	КАЎАЙ	ỹ + й
富士 Fuji	/ɸuɕi/	ФУЦИ	$\phi \rightarrow \Phi$, $\mathfrak{z} \rightarrow \mathfrak{z}$ (merged)
ラーメン rāmen	/ra:men/	РАМЕН	chōonpu → the length bar
寿司 sushi	/suɕi/	СУШИ	$\omega \rightarrow y$ (merged)

German (Standard) — mergers: $\varepsilon:/e/ \rightarrow \bar{e}$

München	/ˈmʏnçən/	МҮНХЬН	ỹ /ʏ/ + хь /ç/ ich-Laut
Goethe	/ˈgø:tə/	ГӨТЕ	long ̄ /ø:/
Zürich	/ˈtsy:ʁɪç/	ЦҮРИХЬ	ц + ỹ
Straße	/ˈʃtʁa:sə/	ШТРАСЕ	шт cluster — native to Bulgarian
Bach	/bax/	БАХ	ach-Laut = native x
schön	/ʃø:n/	ШӨН	̄

Korean (Seoul) — mergers: tense = geminate

서울 Seoul	/sɭ.ul/	СЪУЛ	ㅅ /ɭ/ → ь
김치 kimchi	/kimtɕʰi/	КИМЧЬИ	aspirated чʰ
한글 hangul	/hangwɭ/	ҲАНГЫЛ	h + ы /w/
태권도 taekwondo	/tʰɛkʷando/	ТʰЕККҮЬНДО	tʰ aspiration; tense ɕ = кк
부산 Busan	/pusan/	ПУСАН	lax п

Vietnamese (Hanoi) — mergers: $\mathfrak{b} \mathfrak{d} \rightarrow \mathfrak{b} \mathfrak{d}$; declared loss: 6 tones → 4 marks; length under tone

phở	/fə:ɬɰ/	ФЎ	ъ /ə:/ + tone ̣ (length drops under tone)
Việt Nam	/vjət̚ na:m/	ВЬЕТ НАМ	ье
Hà Nội	/ha: nôi/	ҺА НӨЙ	h + tones
nước	/niék/	НЬГЬК	ы /h/ + rising tone
bánh mì	/báɲ m̥i/	БАҢЬ МИ	implosive $\mathfrak{b} \rightarrow \mathfrak{b}$ (merged)

Turkish (Istanbul) — mergers: $\mathfrak{t} \rightarrow \mathfrak{t}$

Atatürk	/aˈtatyrɕ/	АТАТҮРК	ỹ /y/
---------	------------	---------	-------

Source	IPA	Chuzhditsa	Notes
ırmak	/wɪr'mak/	ЫРМАК	ı /w/ → ы
göl	/jøl/	ГӖЛ	ö /ø/
dağ	/da:/	Дӑ	ğ = pure length → ӑ
teşekkür	/teʃek'kyɾ/	ТЕШЕККӱр	geminate + ӱ
İstanbul	/is'tanbuɫ/	ИСТАНБУЛ	ɫ → л (merged)
Italian (Standard) — mergers: ε ɔ → е о; ʎʎ → ль			
pizza	/'pittsa/	ПИЦЦА	the geminate is written double
zero	/'dze:ro/	СЕРО	s /dz/ — alive in Macedonian, borrowed here
gnocchi	/'ɲɔkki/	НЬОККИ	нь + кк
famiglia	/'fa'miʎʎa/	ФАМИЛЬЯ	ль — palatal geminates degeminate (declared)
spaghetti	/spa'getti/	СПАГЕТТИ	тт
Persian (Tehran) — mergers: ɟ → қ; declared loss: ʔ dropped			
تهران Tehrān	/teh'rɔ:n/	Теһрāн	h between vowels
قم Qom	/gom/	Қом	қ /g/
خدا khodā	/xo'dɔ:/	ХОДӑ	x /x/
شیراز Shirāz	/ʃi:'rɔ:z/	Шйрāз	й + ā
دوغ dugh	/du:g/	Дӱк	غ/ق merged in Tehran → қ
Swahili (Standard) — mergers: implosives → б д			
dhahabu	/ðɑ'habu/	ЗАҺАБУ	з + h (Arabic loans in Swahili)
thelathini	/θela'θini/	ЦЕЛАЦИНИ	double ц
ng'ombe	/ɲombe/	НОМБЕ	initial ɲ — impossible today
ghali	/ɣali/	ГАЛИ	ғ
Kenya	/keɲa/	КЕНЯ	/ɲa/ = ня, the native mechanism
Greek (Standard Modern)			
Θεσσαλονίκη	/θesalo'nici/	ЦЕСАЛОНИКИ	θ → ц
δεν	/ðen/	ЗЕН	ð → з
γάλα	/'ɣala/	ГАЛА	γ → ғ
ευχαριστώ	/efxari'sto/	ЕФХАРИСТО	χ = native x
χάος	/'xaos/	ХАОС	Greek becomes fully writable
Polish (Standard) — mergers: ɕ → ш; tɕ tɕ → ч			
Wrocław	/'vrɔtswaf/	Вроцѡф	ɫ /w/ → ѡ
maąka	/'mɔ̃ka/	МХКА	ą is the old nasal: the yus meets its own descendant

Source	IPA	Chuzhditsa	Notes
pięć	/p'ɛ̃tɕ/	ПѡЧ	ѡ — the font fuses ѡѡ by itself
szczęście	/'ʂtʂɛ̃ɕtɕɛ/	ШЧѡСЪЧѢ	the famous cluster, tamed
Kraków	/'krakuf/	КракуѢ	final devoicing, honestly
Indonesian (Standard)			
orang	/'oraŋ/	ораѢ	Ѣ
bahasa	/ba'hasa/	баѢаса	Ѣ
terima kasih	/tə'rima 'kasih/	тъѢрма касиѢ	schwa → Ѣ, final Ѣ
nyanyi	/ŋaŋi/	няѢѢи	ѢѢ
ngomong	/ŋɔmɔŋ/	ѢомоѢѢ	Ѣ in two positions